

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 35)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 1

. 1

Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

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0 1

. 2

Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

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0 1 . 3

Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

0 1 . 4

Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

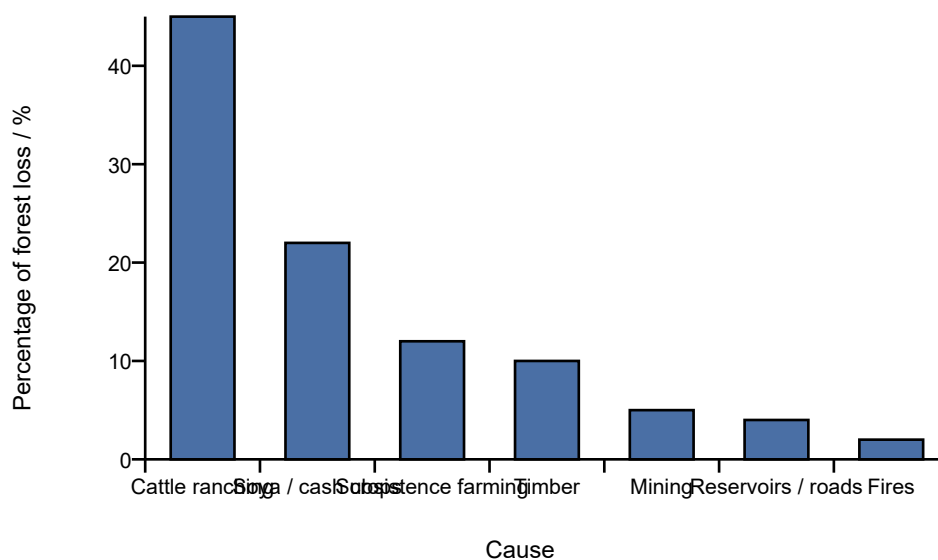
[1 mark]

10

0 2

Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



0 2 . 1

Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

0 2 . 2

Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

0 2 . 3

Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction

HEP reservoirs

Climate change

0 2 . 4

Describe how a debt-for-nature swap works.

[2 marks]

10

03

Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

03

1

Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

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Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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3

State the equation used to forecast a change in population size.

[1 mark]

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0 3 . 4

Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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0 4

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 4 . 1

Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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0 4 . 2

Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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3

0 4 . 3 Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

0 4 . 4 Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

10

0 5 Table 1 lists some techniques used in ecological monitoring.

0 5 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

0	5	.	2
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Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

0	5	.	3
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Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

10

06 . 1

Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

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06 . 2

Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

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06 . 3

Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

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06 . 4

Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

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07 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

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DNA databases and eDNA

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Image recognition software

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Acoustic monitoring / sonograms

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07 . 2

Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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10

08 . 1

Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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08 . 2

Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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08 . 3

Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

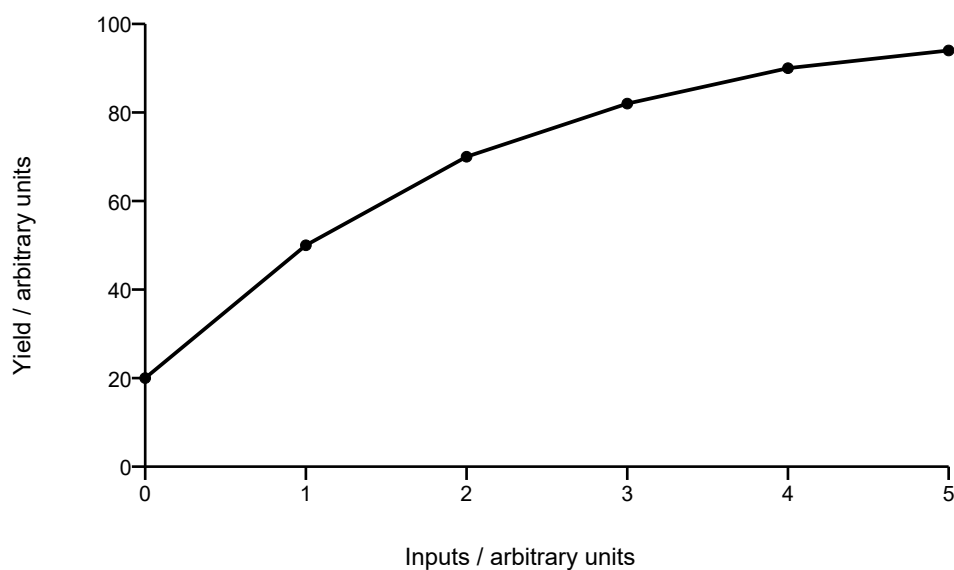
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5

0 9

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1**0 9****. 1**

Describe the relationship shown in Figure 1.

[2 marks]**0 9****. 2**

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]**Answer**

units

0 9**. 3**

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

09 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

09 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

10

10 . 1

Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

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Extra space

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10 . **2** State what is meant by **rewilding**.

[1 mark]

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Write an essay on one of the following topics.

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Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS